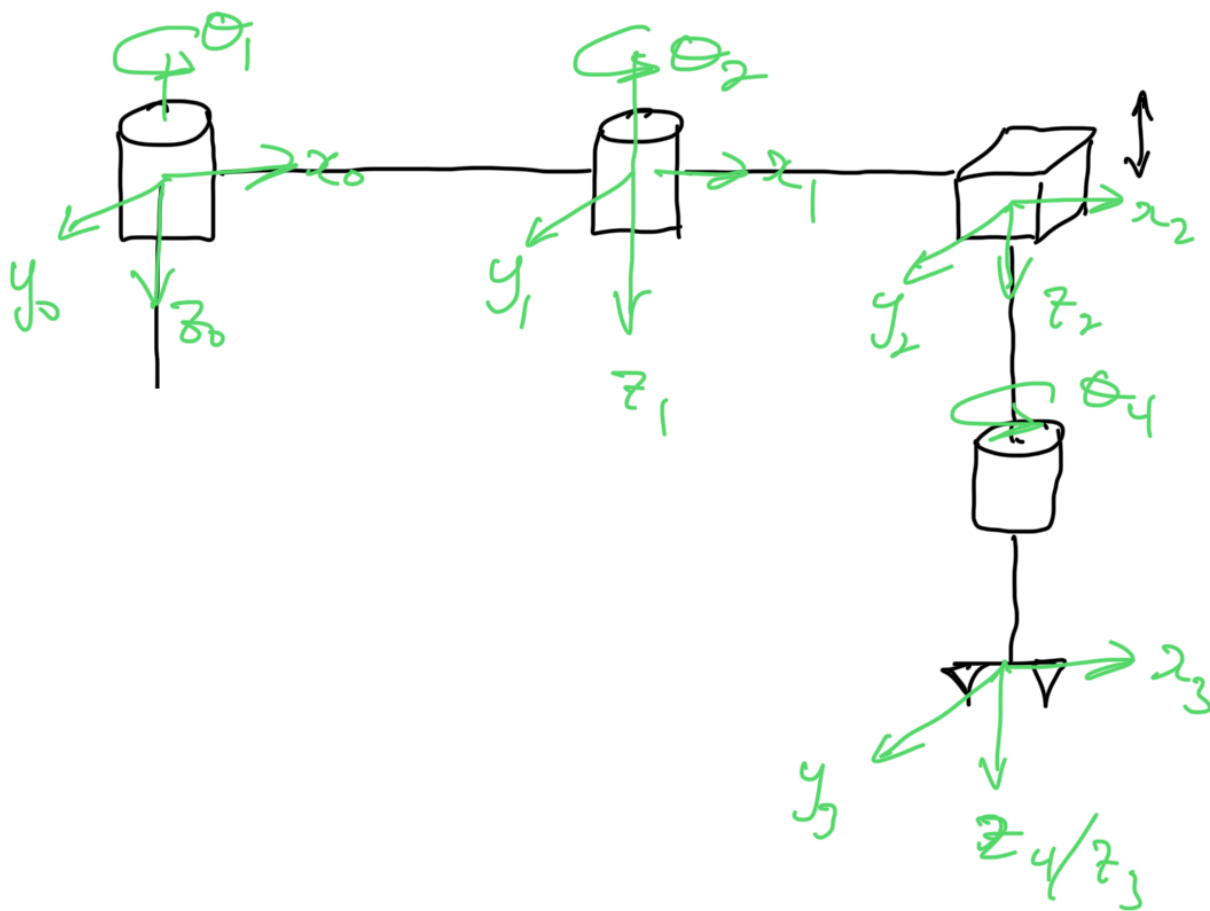




D-H Parameters

Link	a_i	α_i	d_i	θ_i
1	a_1	0	0	θ_1
2	a_2	0	0	θ_2
3	0	0	d_3	0
4	0	0	0	θ_4

Forward Kinematics

$$A_1^0 = \begin{bmatrix} c_1 & -s_1 & 0 & a_1 c_1 \\ s_1 & c_1 & 0 & a_1 s_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_2^1 = \begin{bmatrix} c_2 & -s_2 & 0 & a_2 c_2 \\ s_2 & c_2 & 0 & a_2 s_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_3^2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_4^3 = \begin{bmatrix} c_4 & -s_4 & 0 & 0 \\ s_4 & c_4 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

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$$A_2^0 = A_1^0 A_2^1 = \begin{bmatrix} c_1 & -s_1 & 0 & a_1 c_1 \\ s_1 & c_1 & 0 & a_1 s_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c_2 & -s_2 & 0 & a_2 c_2 \\ s_2 & c_2 & 0 & a_2 s_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} c_{12} & -s_{12} & 0 & a_1 c_1 + a_2 c_{12} \\ s_{12} & c_{12} & 0 & a_1 s_1 + a_2 s_{12} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_3^0 = A_2^0 A_3^2$$

$$\begin{bmatrix} c_{12} & -s_{12} & 0 & a_1 c_1 + a_2 c_{12} \\ s_{12} & c_{12} & 0 & a_1 s_1 + a_2 s_{12} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} c_{12} & -s_{12} & 0 & a_1 c_1 + a_2 c_{12} \\ s_{12} & c_{12} & 0 & a_1 s_1 + a_2 s_{12} \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_4^0 = \begin{bmatrix} c_{124} & -s_{124} & 0 & a_1 c_1 + a_2 c_{124} \\ s_{124} & c_{124} & 0 & a_1 s_1 + a_2 s_{124} \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

(a). Compute Geometric Jacobian

$$J = \begin{bmatrix} J_v \ 3 \times 4 \\ J_\omega \ 3 \times 4 \end{bmatrix}$$

Angular Velocity Jacobian

$$J_\omega = \begin{bmatrix} z_0 & z_1 & 0 & z_3 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}$$

Linear Velocity Jacobian

$$J_v = \begin{bmatrix} z_0 \times (p_2 - p_0) & z_1 \times (p_2 - p_1) & z_2 & z_3 \times (p_2 - p_3) \end{bmatrix}$$

$$z_0 \times (p_2 - p_0) = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \times \begin{bmatrix} a_1 c_1 + a_2 c_{124} \\ a_1 s_1 + a_2 s_{124} \\ d_3 \end{bmatrix}$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & 1 \\ a_1 c_1 + a_2 c_{124} & a_1 s_1 + a_2 s_{124} & d_3 \end{vmatrix}$$

$$= a_1 s_1 + a_2 s_{124} \hat{i} - (a_1 c_1 + a_2 c_{124}) \hat{j} + 0 \hat{k}$$

$$\begin{aligned} \vec{z}_1 \times (\vec{p}_e - \vec{p}_1) &= \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \times \begin{bmatrix} a_2 C_{124} \\ a_2 S_{124} \\ d_3 \end{bmatrix} \\ &= a_2 S_{124} \hat{i} - a_2 C_{124} \hat{j} + 0 \hat{k} \end{aligned}$$

$$\vec{z}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\begin{aligned} \vec{z}_3 \times (\vec{p}_e - \vec{p}_3) &= \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \times \begin{bmatrix} a_2 C_{124} - a_2 C_{12} \\ a_2 S_{124} - a_2 S_{12} \\ d_3 \end{bmatrix} \\ &= a_2 (S_{124} - S_{12}) - a_2 (C_{124} - C_{12}) \hat{j} + 0 \hat{k} \end{aligned}$$

$$\vec{J}_V = \begin{bmatrix} a_1 S_1 + a_2 S_{124} & a_2 S_{124} & 0 & a_2 (S_{124} - S_{12}) \\ -a_1 C_1 - a_2 C_{124} & -a_2 C_{124} & 0 & -a_2 (C_{124} - C_{12}) \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Geometric Jacobian

$$\vec{J} = \begin{bmatrix} a_1 S_1 + a_2 S_{124} & a_2 S_{124} & 0 & a_2 (S_{124} - S_{12}) \\ -a_1 C_1 - a_2 C_{124} & -a_2 C_{124} & 0 & -a_2 (C_{124} - C_{12}) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}$$

(b). Compute Analytical Jacobian

$$\bar{J} = T_A(\phi) \cdot J_A(q)$$

$$\text{where } T_A = \begin{bmatrix} I_3 & 0 \\ 0 & T(\phi) \end{bmatrix}$$

$$\text{and } \phi = \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix}, \quad J_A = \begin{bmatrix} J_p \\ J_\phi \end{bmatrix}$$

Using Roll - Pitch - Yaw (α, β, γ)

$$\begin{aligned} \omega &= \dot{\alpha} z_0 + \dot{\beta} y_1 + \dot{\gamma} x_2 \\ &= \begin{bmatrix} z_0 & y_1 & x_2 \end{bmatrix} \begin{bmatrix} \dot{\alpha} \\ \dot{\beta} \\ \dot{\gamma} \end{bmatrix} \end{aligned}$$

$$R^0 = R_z(\alpha) = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$z_0 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$R^1 = R_z(\alpha) R_{y_1}(\beta)$$

$$= \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \beta & 0 & \sin \beta \\ 0 & 1 & 0 \\ -\sin \beta & 0 & \cos \beta \end{bmatrix}$$

$$= \begin{bmatrix} \cos \alpha \cos \beta & -\sin \alpha & \cos \alpha \sin \beta \\ \sin \alpha \cos \beta & \cos \alpha & \sin \alpha \sin \beta \\ -\sin \beta & 0 & \cos \beta \end{bmatrix}$$

$$y_1 = \begin{bmatrix} -\sin \alpha \\ \cos \alpha \\ 0 \end{bmatrix}$$

$$R^2 = R_z(\alpha) \cdot R_y(\beta) \cdot R_x(\gamma)$$

$$= \begin{bmatrix} \cos \alpha \cos \beta & -\sin \alpha & \cos \alpha \sin \beta \\ \sin \alpha \cos \beta & \cos \alpha & \sin \alpha \sin \beta \\ -\sin \beta & 0 & \cos \beta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \gamma & -\sin \gamma \\ 0 & \sin \gamma & \cos \gamma \end{bmatrix}$$

$$= \begin{bmatrix} \cos \alpha \cos \beta & \cos \alpha \sin \beta \sin \gamma - \sin \alpha \cos \gamma & \cos \alpha \sin \beta \cos \gamma + \sin \alpha \sin \gamma \\ \sin \alpha \cos \beta & \sin \alpha \sin \beta \sin \gamma + \cos \alpha \cos \gamma & \sin \alpha \sin \beta \cos \gamma - \cos \alpha \sin \gamma \\ -\sin \beta & \cos \beta \sin \gamma & \cos \beta \cos \gamma \end{bmatrix}$$

$$\alpha_2 = \begin{bmatrix} \cos \alpha \cos \beta \\ \sin \alpha \cos \beta \\ -\sin \beta \end{bmatrix}$$

$$\therefore T = \begin{bmatrix} 0 & -\sin \alpha & \cos \alpha \cos \beta \\ 0 & \cos \alpha & \sin \alpha \cos \beta \\ 1 & 0 & -\sin \beta \end{bmatrix}$$

$$|T| = \begin{vmatrix} -\sin \alpha & \cos \alpha \cos \beta \\ \cos \alpha & \sin \alpha \cos \beta \end{vmatrix}$$

$$= -\sin^2 \alpha \cos \beta - \cos^2 \alpha \cos \beta = -\cos \beta$$

$$\text{Adj}(T) = C^T = \begin{bmatrix} -\cos\alpha \sin\beta & \sin\alpha \cos\beta & -\cos\alpha \\ -\sin\alpha \sin\beta & -\cos\alpha \cos\beta & -\sin\alpha \\ -\cos\beta & 0 & 0 \end{bmatrix}^T$$

$$T^{-1} = \frac{1}{\det(T)} \cdot \text{Adj}(T) = \begin{bmatrix} \cos\alpha \tan\beta & \sin\alpha \tan\beta & 1 \\ -\sin\alpha & \cos\alpha & 0 \\ \frac{\cos\alpha}{\cos\beta} & \frac{\sin\alpha}{\sin\beta} & 0 \end{bmatrix}$$

$$\text{Since } J_\omega = T \cdot J_\phi$$

$$J_\phi = T^{-1} J_\omega$$

$$\text{for } \cos\beta \neq 0 \Rightarrow \beta \neq \frac{\pi}{2}, \frac{3\pi}{2}$$

The Transform T is only invertible if

$$\beta \neq \left\{ \frac{\pi}{2}, \frac{3\pi}{2} \right\}$$

$$\therefore J_\phi = \begin{bmatrix} \cos\alpha \tan\beta & \sin\alpha \tan\beta & 1 \\ -\sin\alpha & \cos\alpha & 0 \\ \frac{\cos\alpha}{\cos\beta} & \frac{\sin\alpha}{\sin\beta} & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

Analytic Jacobian

$$\therefore \bar{J}_A \equiv \begin{bmatrix} a_1 s_1 + a_2 s_{124} & a_2 s_{124} & 0 & a_2 (s_{124} - s_{12}) \\ -a_1 c_1 - a_2 c_{124} & -a_2 c_{124} & 0 & -a_2 (c_{124} - c_{12}) \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Problem 2

Singularities of 3 Link Planar Arm.

From Class Notes

The Jacobian for the 3-Link Planar Arm is

$$\begin{bmatrix} -a_1 s_1 - a_2 s_{12} - a_3 s_{123} & -a_2 s_{12} - a_3 s_{123} & -a_3 s_{123} \\ a_1 c_1 + a_2 c_{12} + a_3 c_{123} & a_2 c_{12} + a_3 c_{123} & a_3 c_{123} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

Singularities occur when the determinant of the Jacobian is zero. This shows that

By Removing the linearly dependent rows.

$$J^* \equiv \begin{bmatrix} -a_1 s_1 - a_2 s_{12} - a_3 s_{123} & -a_2 s_{12} - a_3 s_{123} & -a_3 s_{123} \\ a_1 c_1 + a_2 c_{12} + a_3 c_{123} & a_2 c_{12} + a_3 c_{123} & a_3 c_{123} \\ 1 & 1 & 1 \end{bmatrix}$$

$$\begin{aligned}
 |\bar{J}^*| &= (-a_1 s_1 - a_2 s_{12} - a_3 s_{123}) (a_2 c_{12} + a_3 c_{123}) \\
 &\quad - (-a_2 s_{12} - a_3 s_{123}) (a_1 c_1 + a_2 c_{12} + a_3 c_{123}) \\
 &\quad - a_3 c_{123} [(-a_1 s_1 - a_2 s_{12} - a_3 s_{123}) - (-a_2 s_{12} - a_3 s_{123})] \\
 &\quad - a_3 s_{123} [(a_1 c_1 + a_2 c_{12} + a_3 c_{123}) - (a_2 c_{12} + a_3 c_{123})] \\
 &= a_2 c_{12} (-a_1 s_1 - \cancel{a_2 s_{12}} - \cancel{a_3 s_{123}}) - \cancel{a_3 c_{123}} (\cancel{a_2 s_{12}} + \cancel{a_3 s_{123}}) \\
 &\quad + a_2 s_{12} (a_1 c_1 + \cancel{a_2 c_{12}} + \cancel{a_3 c_{123}}) + \cancel{a_3 s_{123}} (\cancel{a_2 c_{12}} + \cancel{a_3 c_{123}})
 \end{aligned}$$

$$= -a_1 a_2 s_1 c_{12} + a_1 a_2 c_1 s_{12}$$

$$= a_1 a_2 (c_1 s_{12} - s_1 c_{12})$$

$$= a_1 a_2 s_2$$

$$\therefore |\bar{J}^*| = a_1 a_2 s_2$$

Singularity at $|\bar{J}^*| = 0$

$$\Rightarrow \sin \theta_2 = 0 \Rightarrow \theta = 0, \pi$$

The two angles represents when link 2 & link 1 are collinear resulting in the loss of a DOF.

$\theta = 0$ link 2 fully stretched aligning with link 1

$\Theta = \pi$ link 2 folded back onto link 1